

ECE238L Computer Logic Design

Lab 2: Intro to Structural Design

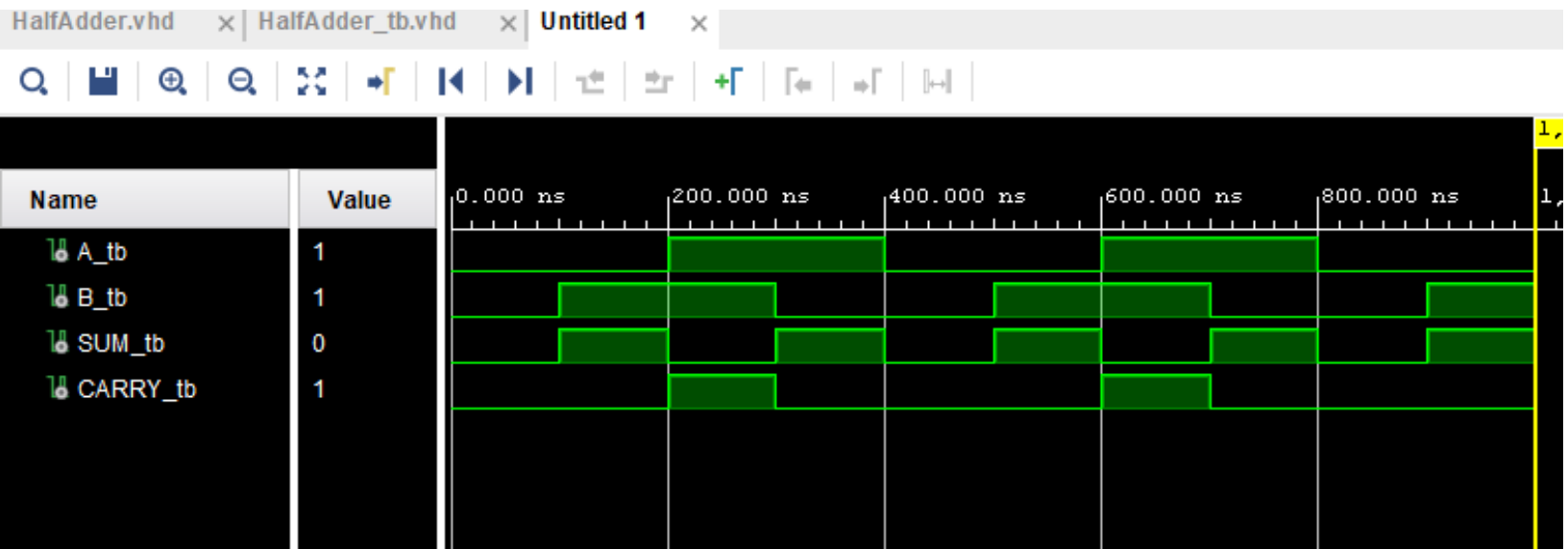
Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

Half Adder VHDL Source Code, Simulation Code and Constraints

```
HalfAdder.vhd
C:/Users/tsalaz25/ECE238L/HalfAdder/HalfAdder.srscs/sources_1/new/HalfAdder.vh
1
2 -- Company:
3 -- Engineer:
4 --
5 -- Create Date: 02/18/2025 01:04:37 PM
6 -- Design Name:
7 -- Module Name: HalfAdder - Behavioral
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12
13 -- Dependencies:
14
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity HalfAdder is
35     Port ( A,B : in STD_LOGIC;
36           SUM : out STD_LOGIC;
37           CARRY : out STD_LOGIC);
38 end HalfAdder;
39
40 architecture Behavioral of HalfAdder is
41
42 begin
43
44 SUM <= (A and not (B)) or (not(A) and B);
45 CARRY <= A and B;
46
47 end Behavioral;
48
HalfAdder_tb.vhd
C:/Users/tsalaz25/ECE238L/HalfAdder/HalfAdder.srscs/sim_1/new/HalfAdder_tb.vhd
12
13 -- Dependencies:
14
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity HalfAdder_tb is
35     -- Port ( );
36 end HalfAdder_tb;
37
38 architecture bench of HalfAdder_tb is
39
40 component HalfAdder
41     Port (A, B : in STD_LOGIC;
42           SUM, CARRY : out STD_LOGIC);
43 end component;
44
45 signal A_tb, B_tb : STD_LOGIC;
46 signal SUM_tb, CARRY_tb : STD_LOGIC;
47
48 begin
49 uut : HalfAdder port map
50     ( A => A_tb,
51       B => B_tb,
52       SUM => SUM_tb,
53       CARRY => CARRY_tb);
54
55 stimulus : process
56     begin
57         A_tb <= '0'; B_tb <= '0'; wait for 100ns;
58         A_tb <= '0'; B_tb <= '1'; wait for 100ns;
59         A_tb <= '1'; B_tb <= '1'; wait for 100ns;
60         A_tb <= '1'; B_tb <= '0'; wait for 100ns;
61     end process;
62 end bench;
HalfAdder_constr.xdc
C:/Users/tsalaz25/ECE238L/HalfAdder/HalfAdder.srscs/constrs_1/new/HalfAdder_constr.xdc
1
2 set_property PACKAGE_PIN J15 [get_ports A]
3 set_property PACKAGE_PIN L16 [get_ports B]
4 set_property PACKAGE_PIN K15 [get_ports CARRY]
5 set_property PACKAGE_PIN H17 [get_ports SUM]
6 set_property IOSTANDARD LVCMOS33 [get_ports A]
7 set_property IOSTANDARD LVCMOS33 [get_ports B]
8 set_property IOSTANDARD LVCMOS33 [get_ports CARRY]
9 set_property IOSTANDARD LVCMOS33 [get_ports SUM]
```

Half Adder Waveform



Half Adder Truth Table and Schematic

Source | Netlist | De x | ? _ □

Internal VREF

- 0.6V
- 0.675V
- 0.75V
- 0.9V

Drop I/O banks on voltages or the "NONE" folder to set/unset Internal VREF.

Cell Prop x | Clock Reg | ? _ □

SUM_OBUF_inst_i_1

I1	I0	O=I0 & !I1 + !I0 & I1
0	0	0
0	1	1
1	0	1
1	1	0

Edit LUT Equation...

Package x | Device x | HalfAdder.vhd x | Schematic x | ? _ □

6 Cells 4 I/O Ports 8 Nets

A IBUF A_IBUF_inst

B IBUF B_IBUF_inst

CARRY_OBUF_inst_i_1

CARRY_OBUF_inst

CARRY

SUM_OBUF_inst_i_1

SUM_OBUF_inst

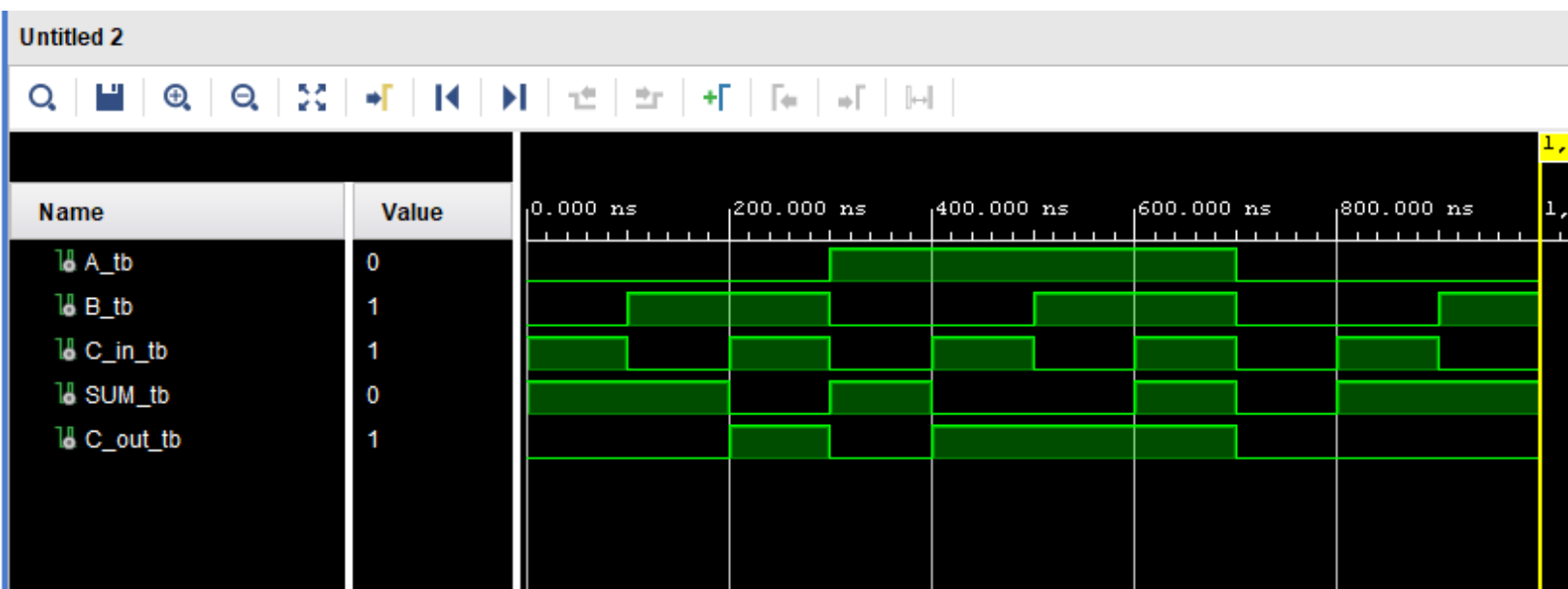
SUM

Truth Table

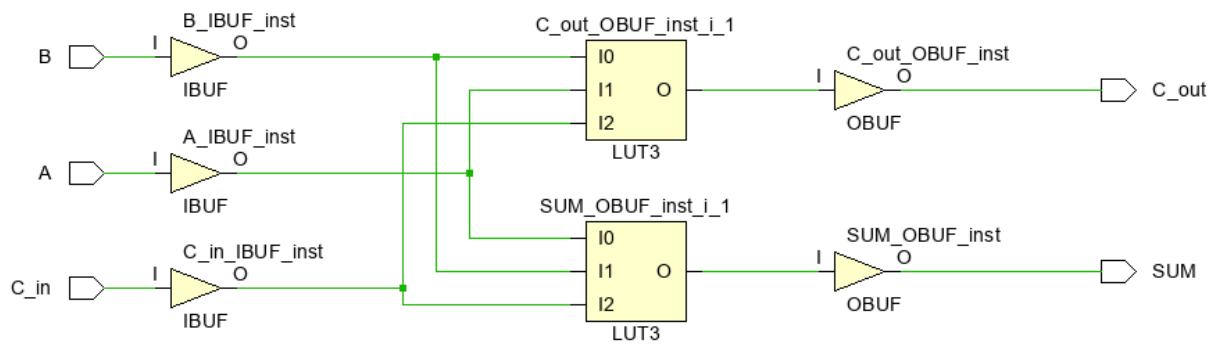
Full Adder VHDL Source Code, Simulation Code, and Constraints

```
FullAdder.vhd
C:/Users/salaz25/ECE238L/FullAdder/FullAdder.srcs/sources_1/new/FullAdder.vhd
7  -- Module Name: FullAdder - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12
13 -- Dependencies:
14
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18
19 -----
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity FullAdder is
35     Port (
36         A : in STD_LOGIC;
37         B : in STD_LOGIC;
38         C_in : in STD_LOGIC;
39         SUM : out STD_LOGIC;
40         C_out : out STD_LOGIC;
41     );
42 end FullAdder;
43
44 architecture Bool_Funct of FullAdder is
45     component HalfAdder
46         Port (
47             A, B : in STD_LOGIC;
48             Sum, Carry : out STD_LOGIC;
49         );
50     end component;
51
52 signal SUM_AB, carry1, carry2 : STD_LOGIC;
53
54 begin
55     ha1 : HalfAdder port map (A => A, B => B, SUM => SUM_AB, Carry => carry1);
56     ha2 : HalfAdder port map (A => SUM_AB, B => C_in, SUM => SUM, Carry => carry2);
57     C_out <= (carry1 or carry2);
58 end Bool_Funct;
59
FullAdder_tb.vhd
C:/Users/salaz25/ECE238L/FullAdder/FullAdder.srcs/sim_1/new/FullAdder_tb.vhd
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity FullAdder_tb is
35     Port (
36         A : in STD_LOGIC;
37         B : in STD_LOGIC;
38         C_in : in STD_LOGIC;
39         SUM : out STD_LOGIC;
40         C_out : out STD_LOGIC;
41     );
42 end FullAdder_tb;
43
44 architecture Behavioral of FullAdder_tb is
45     component FullAdder
46         Port (
47             A, B, C_in : in STD_LOGIC;
48             SUM, C_out : out STD_LOGIC;
49         );
50     end component;
51
52 signal A_tb, B_tb, C_in_tb : STD_LOGIC;
53 signal SUM_tb, C_out_tb : STD_LOGIC;
54
55 begin
56     uut : FullAdder port map
57     (
58         A => A_tb,
59         B => B_tb,
60         C_in => C_in_tb,
61         SUM => SUM_tb,
62         C_out => C_out_tb);
63
64 stimulus : process
65 begin
66     A_tb <= '0'; B_tb <= '0'; C_in_tb <= '1'; wait for 100ns;
67     A_tb <= '0'; B_tb <= '1'; C_in_tb <= '0'; wait for 100ns;
68     A_tb <= '0'; B_tb <= '1'; C_in_tb <= '1'; wait for 100ns;
69     A_tb <= '1'; B_tb <= '0'; C_in_tb <= '0'; wait for 100ns;
70     A_tb <= '1'; B_tb <= '0'; C_in_tb <= '1'; wait for 100ns;
71     A_tb <= '1'; B_tb <= '1'; C_in_tb <= '0'; wait for 100ns;
72     A_tb <= '1'; B_tb <= '1'; C_in_tb <= '1'; wait for 100ns;
73     A_tb <= '0'; B_tb <= '0'; C_in_tb <= '0'; wait for 100ns;
74 end process;
75 end Behavioral;
76
FullAdder_constr.xdc
C:/Users/salaz25/ECE238L/FullAdder/FullAdder.srcs/constrs_1/new/FullAdder
1  set_property IOSTANDARD LVCMOS33 [get_ports A]
2  set_property IOSTANDARD LVCMOS33 [get_ports B]
3  set_property IOSTANDARD LVCMOS33 [get_ports C_in]
4  set_property IOSTANDARD LVCMOS33 [get_ports C_out]
5  set_property IOSTANDARD LVCMOS33 [get_ports SUM]
6  set_property PACKAGE_PIN J15 [get_ports A]
7  set_property PACKAGE_PIN L16 [get_ports B]
8  set_property PACKAGE_PIN M16 [get_ports C_in]
9  set_property PACKAGE_PIN H17 [get_ports C_out]
10 set_property PACKAGE_PIN K15 [get_ports SUM]
```

Full Adder Waveform



Full Adder Truth Table and Schematic



IP INTEGRATOR

0.75V
0.9V

Cell Properties

C_{out}_OBUF_inst_i_1

I2	I1	I0	O=I0 & I1 + I0 & I2 + I1 & I2
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Edit LUT Equation...

er Nets Cell Pins Truth Table

4-Bit Adder VHDL Source Code, Simulation Code, and Constraints

```
RippleCarryAdder.vhd
C:/Users/salaz25/ECE238L/RippleCarryAdder/RippleCarryAdder.srcs/sources_1/new/RippleCarryAdder.vhd

8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23
24 -- Uncomment the following library declaration if using
25 -- arithmetic functions with Signed or Unsigned values
26 --use IEEE.NUMERIC_STD.ALL;
27
28 -- Uncomment the following library declaration if instantiating
29 -- any Xilinx leaf cells in this code.
30 --library UNISIM;
31 --use UNISIM.VComponents.all;
32
33 entity RippleCarryAdder is
34   Port (A,B : in STD_LOGIC_VECTOR (3 downto 0);
35         SUM : out STD_LOGIC_VECTOR (3 downto 0);
36         Cin : in STD_LOGIC;
37         Cout : out STD_LOGIC);
38 end RippleCarryAdder;
39
40 architecture Behavioral of RippleCarryAdder is
41   component FullAdder
42     Port ( A,B,C_in : in STD_LOGIC;
43           SUM, C_out: out STD_LOGIC);
44   end component;
45
46   signal C: STD_LOGIC_VECTOR (2 downto 0);
47
48 begin
49   fa1: FullAdder port map (A => A(0), B => B(0), C_in => Cin, Sum => Sum(0), C_out => C(0));
50   fa2: FullAdder port map (A => A(1), B => B(1), C_in => C(0), Sum => Sum(1), C_out => C(1));
51   fa3: FullAdder port map (A => A(2), B => B(2), C_in => C(1), Sum => Sum(2), C_out => C(2));
52   fa4: FullAdder port map (A => A(3), B => B(3), C_in => C(2), Sum => Sum(3), C_out => Cout);
53
54 end Behavioral;
```

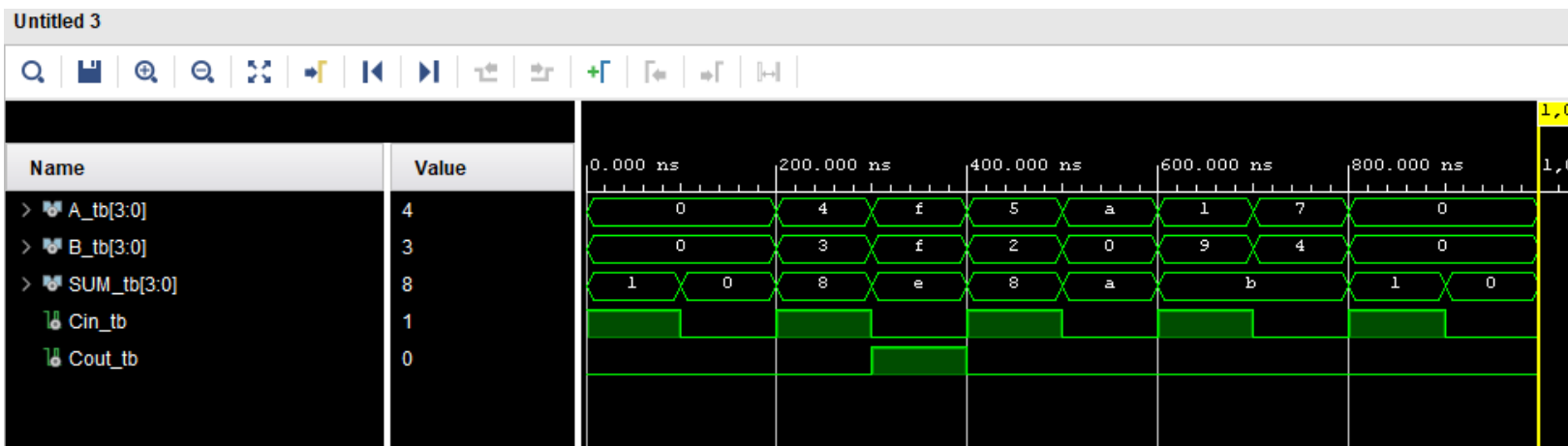
```
RippleCarryAdder_tb.vhd
C:/Users/salaz25/ECE238L/RippleCarryAdder/RippleCarryAdder.srcs/sim_1/new/RippleCarryAdder_tb.vhd

25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity RippleCarryAdder_tb is
35   -- Port ( );
36 end RippleCarryAdder_tb;
37
38 architecture Behavioral of RippleCarryAdder_tb is
39   component RippleCarryAdder
40     Port (A, B : in STD_LOGIC_VECTOR (3 downto 0);
41           SUM : out STD_LOGIC_VECTOR (3 downto 0);
42           Cin : in STD_LOGIC;
43           Cout : out STD_LOGIC);
44   end component;
45
46   signal A_tb, B_tb : STD_LOGIC_VECTOR (3 downto 0);
47   signal SUM_tb : STD_LOGIC_VECTOR (3 downto 0);
48   signal Cin_tb : STD_LOGIC;
49   signal Cout_tb : STD_LOGIC;
50
51 begin
52   uut : RippleCarryAdder port map (
53     A => A_tb,
54     B => B_tb,
55     SUM => SUM_tb,
56     Cin => Cin_tb,
57     Cout => Cout_tb);
58
59   stimulus: process
60   begin
61     A_tb <= "0000"; B_tb <= "0000"; Cin_tb <= '1'; wait for 100ns;
62     A_tb <= "0000"; B_tb <= "0000"; Cin_tb <= '0'; wait for 100ns;
63     A_tb <= "0100"; B_tb <= "0011"; Cin_tb <= '1'; wait for 100ns;
64     A_tb <= "1111"; B_tb <= "1111"; Cin_tb <= '0'; wait for 100ns;
65     A_tb <= "0101"; B_tb <= "0010"; Cin_tb <= '1'; wait for 100ns;
66     A_tb <= "1010"; B_tb <= "0000"; Cin_tb <= '0'; wait for 100ns;
67     A_tb <= "0001"; B_tb <= "1001"; Cin_tb <= '1'; wait for 100ns;
68     A_tb <= "0111"; B_tb <= "0100"; Cin_tb <= '0'; wait for 100ns;
69   end process;
70 end Behavioral;
```

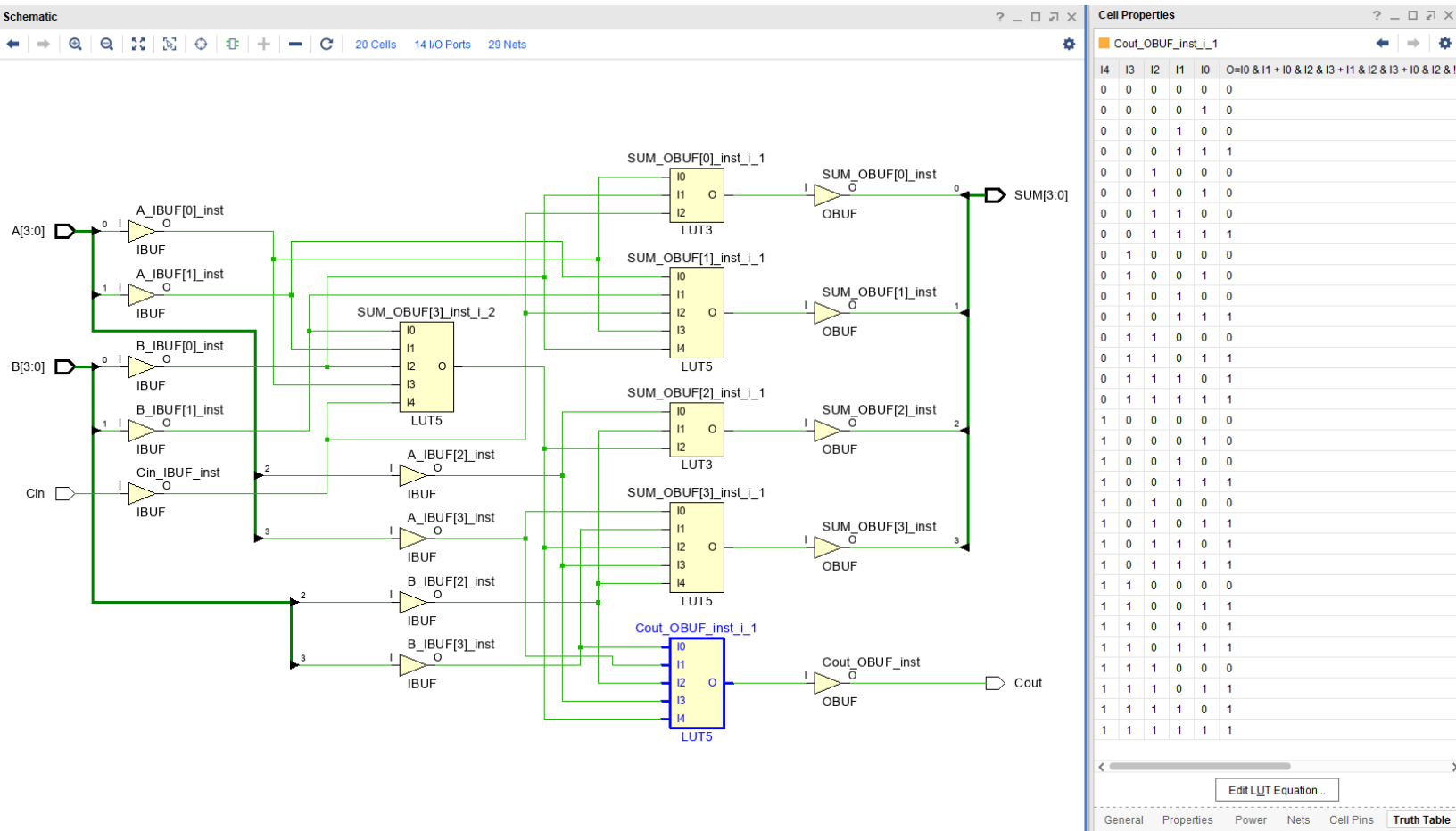
```
RippleCarryAdder_constr.xdc
sersts/salaz25/ECE238L/RippleCarryAdder/RippleCarryAdder.srcs/constr_1/new/RippleCarryAdder_constr.xdc

1 set_property IOSTANDARD LVCMOS33 [get_ports {A[3]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {A[2]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {A[1]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {A[0]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {B[3]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {B[2]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {B[1]}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {B[0]}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {SUM[3]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {SUM[2]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {SUM[1]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {SUM[0]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Cin}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Cout}]
15 set_property PACKAGE_PIN J15 [get_ports {A[0]}]
16 set_property PACKAGE_PIN L16 [get_ports {A[1]}]
17 set_property PACKAGE_PIN M13 [get_ports {A[2]}]
18 set_property PACKAGE_PIN R15 [get_ports {A[3]}]
19 set_property PACKAGE_PIN R17 [get_ports {B[0]}]
20 set_property PACKAGE_PIN T18 [get_ports {B[1]}]
21 set_property PACKAGE_PIN U18 [get_ports {B[2]}]
22 set_property PACKAGE_PIN R13 [get_ports {B[3]}]
23 set_property PACKAGE_PIN U8 [get_ports {Cin}]
24 set_property PACKAGE_PIN V17 [get_ports {Cout}]
25 set_property PACKAGE_PIN H17 [get_ports {SUM[0]}]
26 set_property PACKAGE_PIN K15 [get_ports {SUM[1]}]
27 set_property PACKAGE_PIN J13 [get_ports {SUM[2]}]
28 set_property PACKAGE_PIN N14 [get_ports {SUM[3]}]
29
```

4-Bit Adder Waveform



4-Bit Adder Truth Table and Schematic



Summary:

This Lab was very involved but exemplified important concepts in VHDL. We first had to implement the Half Adder which was used to calculate the 'sum' and the 'carry' to then build the Full Adder using the Half Adder. I think this was the best way to show how we can implement Hierarchical Structure.

The 4-Bit Adder used the previous Adders in its implementation, and while it was more complex, the 4-Bit Adder was not too hard to complete.

Problems:

The syntax in this lab was a bit harder since the elements in different hierarchies have the same name. I also struggled to find add constraints to the subsequent projects, but it was good practice.

Hints:

Be sure to make sure that the Component in the Higher 'class' matches the Entity in the lower (ie 4-Bit Component of Full Adder matches the Full Adder Entity). This gave me some issues.

Recommendations:

NA